

The screenshot displays the Cisco Packet Tracer interface. The main workspace shows a network topology with five 2811 routers (Hammad1-5) and various switches and PCs. The routers are interconnected via serial links. The interface includes a top menu bar, a toolbar, and a bottom status bar with a Realtime log.

Network Topology:

- Router 2811 Hammad1:** Connected to 2811 Hammad5 via Serial 0/3/0 to Fa0/0.
- Router 2811 Hammad2:** Connected to 2811 Hammad3 via Serial 0/3/0 to Fa0/0.
- Router 2811 Hammad3:** Connected to 2811 Hammad4 via Serial 0/3/0 to Fa0/0.
- Router 2811 Hammad4:** Connected to 2811 Hammad8 via Serial 0/3/0 to Fa0/0.
- Router 2811 Hammad5:** Connected to 2811 Hammad1 via Serial 0/3/0 to Fa0/0.

Other Components:

- Switches:** 2960-24TT Switch0, 2960-24TT Switch1, 2960-24TT Switch2, 2960-24TT Switch3, 2960-24TT Switch4.
- PCs:** PC1-PC19.

Bottom Status Bar:

Time: 00:58:35 Power Cycle Devices Fast forward Time

Realtime Log:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
●	Successful	PC17	PC3	ICMP	Green	0.000	N	8	(edit)	(delete)
●	Successful	PC18	PC10	ICMP	Green	0.000	N	9	(edit)	(delete)
●	Successful	PC8	PC2	ICMP	Green	0.000	N	10	(edit)	(delete)

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC5	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	PC14	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC1	PC18	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC1	PC10	ICMP		0.000	N	3	(edit)	(delete)
	Successful	PC7	PC3	ICMP		0.000	N	4	(edit)	(delete)
	Successful	PC6	PC13	ICMP		0.000	N	5	(edit)	(delete)
	Successful	PC5	PC9	ICMP		0.000	N	6	(edit)	(delete)
	Successful	PC4	PC19	ICMP		0.000	N	7	(edit)	(delete)
	Successful	PC17	PC3	ICMP		0.000	N	8	(edit)	(delete)
	Successful	PC18	PC10	ICMP		0.000	N	9	(edit)	(delete)
	Successful	PC8	PC2	ICMP		0.000	N	10	(edit)	(delete)
	Successful	PC11	PC18	ICMP		0.000	N	11	(edit)	(delete)
	Successful	PC8	PC13	ICMP		0.000	N	12	(edit)	(delete)
	Successful	PC6	PC17	ICMP		0.000	N	13	(edit)	(delete)

(Hammad1) CLI command

 Hammad1

Physical Config CLI

IOS C

```
Compiled Wed 18-Jul-07 06:21 by pt_rel_team
```

```
--- System Configuration Dialog ---
```

```
Continue with configuration dialog? [yes/no]: n
```

```
Press RETURN to get started!
```

```
Router>en
```

```
Router#config
```

```
Configuring from terminal, memory, or network [terminal]?
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
Router(config)#exit
```

```
Router#
```

```
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#en
```

```
Router#config t
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
Router(config)#hostname Hammadullah15398
```

```
Hammadullah15398(config)#int fa0/0
```

```
Hammadullah15398(config-if)#ip address 192.168.1.1 255.255.255.0
```

```
Hammadullah15398(config-if)#no shutdown
```

```
Hammadullah15398(config-if)#
```

```
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

```
Hammadullah15398(config-if)#int Se0/3/0
```

```
Hammadullah15398(config-if)#ip address 10.0.0.1 255.255.255.252
```

```
Hammadullah15398(config-if)#clock rate 64000
```

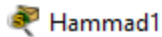
```
Hammadullah15398(config-if)#no shutdown
```

```
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
```

```
Hammadullah15398(config-if)#
```

```
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
```



Physical Config CLI

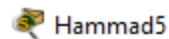
```
Hammadullah15398>en
Hammadullah15398#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Hammadullah15398(config)#router rip
Hammadullah15398(config-router)#version 2
Hammadullah15398(config-router)#no auto-summary
      ^
% Invalid input detected at '^' marker.

Hammadullah15398(config-router)#no auto-summary
Hammadullah15398(config-router)#network 10.0.0.0
Hammadullah15398(config-router)#network 20.0.0.0
Hammadullah15398(config-router)#network 30.0.0.0
Hammadullah15398(config-router)#network 40.0.0.0
Hammadullah15398(config-router)#network 192.168.1.0
Hammadullah15398(config-router)#network 192.168.2.0
Hammadullah15398(config-router)#network 192.168.3.0
Hammadullah15398(config-router)#network 192.168.4.0
Hammadullah15398(config-router)#network 192.168.5.0
Hammadullah15398(config-router)#end
Hammadullah15398#
%SYS-5-CONFIG_I: Configured from console by console

Hammadullah15398#wr
Building configuration...
[OK]
Hammadullah15398#
```

Hammadullah15398 con0 is now available

(Hammad5) CLI command



Physical Config CLI

```
System Configuration Dialog

Continue with configuration dialog? [yes/no]: n

Press RETURN to get started!

Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Hammadullah15398
Hammadullah15398(config)#int fa0/0
Hammadullah15398(config-if)#ip address 192.168.2.1 255.255.255.0
Hammadullah15398(config-if)#no shutdown

Hammadullah15398(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to u

Hammadullah15398(config-if)#int se0/3/0
Hammadullah15398(config-if)#ip address 10.0.0.2 255.255.255.252
Hammadullah15398(config-if)#no shutdown

Hammadullah15398(config-if)#
%LINK-5-CHANGED: Interface Serial10/3/0, changed state to up

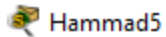
Hammadullah15398(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial10/3/0, changed state to up

Hammadullah15398(config-if)#int se0/3/1
Hammadullah15398(config-if)#ip address 20.0.0.1 255.255.255.252
Hammadullah15398(config-if)#clock rate 64000
Hammadullah15398(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial10/3/1, changed state to down
Hammadullah15398(config-if)#
%LINK-5-CHANGED: Interface Serial10/3/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial10/3/1, changed state to up

Hammadullah15398 con0 is now available
```

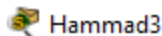


Physical Config CLI

```
Hammadullah15398>en
Hammadullah15398#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Hammadullah15398(config)#router rip
Hammadullah15398(config-router)#version 2
Hammadullah15398(config-router)#no auto-summary
Hammadullah15398(config-router)#network 10.0.0.0
Hammadullah15398(config-router)#network 20.0.0.0
Hammadullah15398(config-router)#network 30.0.0.0
Hammadullah15398(config-router)#network 40.0.0.0
Hammadullah15398(config-router)#network 192.168.1.0
Hammadullah15398(config-router)#network 192.168.2.0
Hammadullah15398(config-router)#network 192.168.3.0
Hammadullah15398(config-router)#network 192.168.4.0
Hammadullah15398(config-router)#network 192.168.5.0
Hammadullah15398(config-router)#end
Hammadullah15398#
%SYS-5-CONFIG_I: Configured from console by console

Hammadullah15398#wr
Building configuration...
[OK]
Hammadullah15398#
```

Hammad3(CLI Command)



Physical Config CLI

--- System Configuration Dialog ---

Continue with configuration dialog? [yes/no]: n

Press RETURN to get started!

Router>en

Router#config t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#hostname Hammadullah15398

Hammadullah15398(config)#int fa0/0

Hammadullah15398(config-if)#ip address 192.168.3.1 255.255.255.0

Hammadullah15398(config-if)#no shutdown

Hammadullah15398(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Hammadullah15398(config-if)#int se0/3/0

Hammadullah15398(config-if)#ip address 20.0.0.2 255.255.255.252

Hammadullah15398(config-if)#no shutdown

Hammadullah15398(config-if)#

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

Hammadullah15398(config-if)#int se0/3/1

Hammadullah15398(config-if)#

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Hammadullah15398(config-if)#int se0/3/1

Hammadullah15398(config-if)#ip address 30.0.0.1 255.255.255.252

Hammadullah15398(config-if)#clock rate 64000

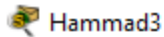
Hammadullah15398(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down

Hammadullah15398(config-if)#

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up



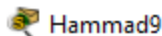
Physical Config CLI

Press RETURN to get started.

```
Hammadullah15398>en
Hammadullah15398#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Hammadullah15398(config)#router rip
Hammadullah15398(config-router)#version 2
Hammadullah15398(config-router)#no auto-summary
Hammadullah15398(config-router)#network 10.0.0.0
Hammadullah15398(config-router)#network 20.0.0.0
Hammadullah15398(config-router)#network 30.0.0.0
Hammadullah15398(config-router)#network 40.0.0.0
Hammadullah15398(config-router)#network 192.168.1.0
Hammadullah15398(config-router)#network 192.168.2.0
Hammadullah15398(config-router)#network 192.168.3.0
Hammadullah15398(config-router)#network 192.168.4.0
Hammadullah15398(config-router)#network 192.168.5.0
Hammadullah15398(config-router)#end
Hammadullah15398#
%SYS-5-CONFIG_I: Configured from console by console

Hammadullah15398#wr
Building configuration...
[OK]
Hammadullah15398#
```

Hammad9(CLI Command)



Physical Config CLI

Continue with configuration dialog? [yes/no]: n

Press RETURN to get started!

Router>en

Router#config t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#hostname Hammadullah15398

Hammadullah15398(config)#int fa0/0

Hammadullah15398(config-if)#ip address 192.168.4.1 255.255.255.0

Hammadullah15398(config-if)#no shutdown

Hammadullah15398(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Hammadullah15398(config-if)#int se0/3/0

Hammadullah15398(config-if)#ip address 30.0.0.2 255.255.255.252

Hammadullah15398(config-if)#no shutdown

Hammadullah15398(config-if)#

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

Hammadullah15398(config-if)#int se0/3/1

Hammadullah15398(config-if)#ip address

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

% Incomplete command.

Hammadullah15398(config-if)#int se0/3/1

Hammadullah15398(config-if)#ip address 40.0.0.1 255.255.255.252

Hammadullah15398(config-if)#clock rate 64000

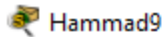
Hammadullah15398(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down

Hammadullah15398(config-if)#

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up



Physical Config CLI

Press RETURN to get started.

```
Hammadullah15398>en
Hammadullah15398#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Hammadullah15398(config)#router rip
Hammadullah15398(config-router)#version 2
Hammadullah15398(config-router)#no auto-summary
Hammadullah15398(config-router)#network 10.0.0.0
Hammadullah15398(config-router)#network 20.0.0.0
Hammadullah15398(config-router)#network 30.0.0.0
Hammadullah15398(config-router)#network 40.0.0.0
Hammadullah15398(config-router)#network 192.168.1.0
Hammadullah15398(config-router)#network 192.168.2.0
Hammadullah15398(config-router)#network 192.168.3.0
Hammadullah15398(config-router)#network 192.168.4.0
Hammadullah15398(config-router)#network 192.168.5.0
Hammadullah15398(config-router)#end
Hammadullah15398#
%SYS-5-CONFIG_I: Configured from console by console

Hammadullah15398#wr
Building configuration...
[OK]
Hammadullah15398#
```

Hammadullah15398 con0 is now available

Hammad8(Cli Command)

 Hamad8

Physical Config CLI

IOS Com

```
62720K bytes of ATA CompactFlash (Read/Write)
Cisco IOS Software, 2800 Software (C2800NM-ADVIPSERVICESK9-M), Version 12.4(15)T1, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 06:21 by pt_rel_team
```

```
--- System Configuration Dialog ---
```

```
Continue with configuration dialog? [yes/no]: n
```

```
Press RETURN to get started!
```

```
Router>en
```

```
Router#config t
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
Router(config)#hostname Hammadullah15398
```

```
Hammadullah15398(config)#int fa0/0
```

```
Hammadullah15398(config-if)#ip address 192.168.5.1 255.255.255.0
```

```
Hammadullah15398(config-if)#no shutdown
```

```
Hammadullah15398(config-if)#
```

```
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

```
Hammadullah15398(config-if)#int se0/3/0
```

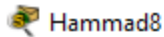
```
Hammadullah15398(config-if)#ip address 40.0.0.2 255.255.255.252
```

```
Hammadullah15398(config-if)#no shutdown
```

```
Hammadullah15398(config-if)#
```

```
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
```



Hammad8

Physical Config CLI

```
Hamadullah15398>en
Hamadullah15398#config t
Enter configuration commands, one per line. End with CNTL/Z.
Hamadullah15398(config)#router rip
Hamadullah15398(config-router)#version 2
Hamadullah15398(config-router)#no auto-summary
Hamadullah15398(config-router)#network 10.0.0.0
Hamadullah15398(config-router)#network 20.0.0.0
Hamadullah15398(config-router)#network 30.0.0.0
Hamadullah15398(config-router)#network 40.0.0.0
Hamadullah15398(config-router)#network 192.168.1.0
Hamadullah15398(config-router)#network 192.168.2.0
Hamadullah15398(config-router)#network 192.168.3.0
Hamadullah15398(config-router)#network 192.168.4.0
Hamadullah15398(config-router)#network 192.168.5.0
Hamadullah15398(config-router)#end
Hamadullah15398#
%SYS-5-CONFIG_I: Configured from console by console

Hamadullah15398#wr
Building configuration...
[OK]
Hamadullah15398#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/30 is subnetted, 1 subnets
R       10.0.0.0 [120/3] via 40.0.0.1, 00:00:08, Serial0/3/0
    20.0.0.0/30 is subnetted, 1 subnets
R       20.0.0.0 [120/2] via 40.0.0.1, 00:00:08, Serial0/3/0
    30.0.0.0/30 is subnetted, 1 subnets
R       30.0.0.0 [120/1] via 40.0.0.1, 00:00:08, Serial0/3/0
    40.0.0.0/30 is subnetted, 1 subnets
C       40.0.0.0 is directly connected, Serial0/3/0
R       192.168.1.0/24 [120/4] via 40.0.0.1, 00:00:08, Serial0/3/0
R       192.168.2.0/24 [120/3] via 40.0.0.1, 00:00:08, Serial0/3/0
R       192.168.3.0/24 [120/2] via 40.0.0.1, 00:00:08, Serial0/3/0
R       192.168.4.0/24 [120/1] via 40.0.0.1, 00:00:08, Serial0/3/0
--More--
```